LAB 01

→ Objectives

1) Getting started with the Linux Operating System (login, execute simple commands, edit file, transfer files)

☒ Write a simple "Hello World" program in C, compile and execute the program

   a) gcc <file name> -o my_program
   b) ./my_program

→ Opening a Terminal

Although Linux has a fancy graphical user interface and it works pretty much like other operating systems such as Windows and OS X, we will focus on the command line interface running bash shell.

Command Line Interface Running Bash Shell == Terminal

A terminal is a text-based interface where you can enter commands and get feedback as text.

Opening the Terminal:

MAC: Applications → Utilities → Terminal

LINUX: Applications → System or Applications → Utilities (or, just 'right-click' on the desktop and click "open in terminal")

→ GitHub Codespaces Setup

☒ Make a small change to the README.md document

☒ Create a text file, to initiate the first commit. Make sure the file is "pushed" properly.

   a) git add <file name>
   b) git commit -m "Your commit message here"
   c) git push origin "HEAD or main"

☒ Once done with codespace, stop the codespace.

   a) In the bottom left corner of the screen, you should see a button that states: "Codespaces: <randomly generated name>
   b) Click this button and click stop codespace in the pop up menu

→ Lab Exercise

- ☒ Open a terminal
- ☒ Create a directory called CS-532
- ☒ Change to the CS-532 directory
- ☒ Create a new directory inside CS-532 and name it Lab01
- ☒ Change to the Lab01 directory
- ☒ Find the current directory using the command “pwd”
- ☒ Create a new file using the touch command by entering “touch filename.txt”
- ☒ List the contents of the Lab01 directory
- ☒ Rename the file to new_file1
- ☒ List the contents of the Lab01 directory (again)
- ☒ Change to the CS-532 directory
- ☒ Try to delete the directory Lab01. Do you get an error? What is the error message?
    - error message: **rmdir: failed to remove ‘Lab01’: Directory not empty**
- ☒ Delete the file new_file1 in the directory Lab01
- ☒ Try to delete the directory Lab01. Do you still get an error?
- ☒ Use the “man” command to find out more about the touch command

NOTE: when using “rm” command, to ensure the file removed is removed not only in GitHub Codespaces but in the GitHub repository as well … follow these steps:

I) rm “filename.txt”
II) git commit -a -m “Deleted unnecessary file”
III) git push origin main

→ Editing Text-Based Documents in Linux

As the terminal is text-based, editing documents in a terminal could be challenging. There are several well-known text editors in a Linux environment such as vi, emacs, nano, etc. We will use nano in this lab to create a new file, type some text and save the file.

- ☒ Enter “nano file1” at the command prompt
- ☒ Enter the following text:

- you can use the arrow keys to move backward/forward/up/down and backspace to delete a character)

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The Student Conduct Code promotes honesty, integrity, accountability, rights and responsibilities expected of students consisted with the core missions of the University of Alabama at Birmingham. This Code describes the standards or behavior for all students, and outlines student's rights, responsibilities, and the campus processes for adjudicating alleged violations.

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- ☒ Save the file by entering "Control-O"
  - you will be prompted with "File Name to Write: file1", hit enter)
- ☒ Exit the editor by entering "Control-X"
- ☒ View the contents of the file using one of the following commands:
  1) more "filename.txt"
  2) less "filename.txt"
  3) cat "filename.txt"

What is the difference between the three commands?

cat → quickly print the entire file to your screen

less → scroll through large files both forwards and backwards

more → read files page-by-page scrolling only downwards

NOTE: You can use the editor to make changes to the file you just created or create a new file like we just did.

→ Transferring Files Between your Local Computer and CS Linux Systems

NOTE: moat.cis.uab.edu is no longer an active server, use GitHub cloud accessibility with Codespaces.

Mac and Linux users can use the **scp** command to copy files from your local computer to the CS Linux systems

- ☒ Use the following example:

for files: scp mylocalfile.txt *blazerID*@moat.cis.uad.edu:CS532/Lab1

for folders: scp -r anyDirectory *blazerID*@moat.cis.uab.edu:CS532/

→ tried scp conductCode bbelden@moat.cis.uab.edu:CS532/Lab1

moat.cis.uab.edu is most likely a department-managed Linux server operated by the Computer Science Department at UAB.

the server is probably where:

- Your CS course accounts are hosted.
- You log in to complete programming assignments.
- Your course directories are stored.
- Instructors and teaching assistants can access and grade your work.

This will copy the file mylocalfile.txt in the current directory of your local computer to the CS532/Lab1 directory on the CS Linux system.

If you would like to copy a file from the CS Linux system to your local computer, you must execute the following command on your local computer:

☒ Copy a file from CS Linux system to local computer

scp *blazerID*@moat.cis.uab.edu:CS532/Lab1/new_file1 .

This will copy the file new_file1 in the directory CS532/Lab1 on the CS Linux system to the current directory of your local computer.

→ Lab Assignment #1

Write a C program to check whether given number is prime or not by performing following steps:

☒ Define an integer variable as given *given_number*

☒ Use the C function *scanf* to read the integer variable *given_number*

☒ Use the conditional statement to find out *given_number* is prime or not prime

☒ Print the final output as *The number is prime* or *The number is not prime*

Compile and test the program on CS Linux systems and upload the C source code in the lab assignment submission section.

Terminal commands:

1) cd Lab01
2) touch isPrime.c
3) gcc isPrime.c -o isPrime
4) ./isPrime

→ zipping in GitHub Codespaces

- make sure current location - /workspaces/CS-532
- "git archive –format=zip HEAD Lab01 -o Lab01.zip"
- mv Lab01.zip /Lab01
- right click, download
- drag and drop to desktop